Talk with Hipp, MacSorley, et all
on Mpy Time

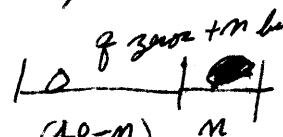
June 20, 1958

assemble Mpier in [redacted] 6 bytes fixed (n bits)
M card 6 bytes (n bits)
housekeeping (sign byte, etc) 2

mpy & check = ~~[redacted]~~
$n(.4) + 1$ formula for lawspeed

---

there are $\frac{n}{3}$ adds in skipping mpies. ← Note.



$\frac{n}{3}$ + of bits shift

.45 = add time

$= .54 \frac{n}{3}$

$.45\frac{n}{3} + 48 - n = .18n$

$48 - n = .03n$

$1.03n = 48$

$\frac{n+m}{8} + .4n + 15$

VFL Mpy = assemble time + mpy time

will be 7–8 usec
10–12

Start with low-order bit

use fixed jump max of 4. (look at 5 bits for testing)

must take $\frac{48}{3} = 16.$ instead of $.4n$

to replace prod.

Should be: ? $\left[15 + \frac{n+m}{8}\right] 0.45 + 7.0\ \mu s$

fl. mpy.

to assemble n & m (always take full size)

div time: $\frac{9}{3}$

8 bit = 14.5 not 9.5

48 bit = 19.15 20.71

as listed

add: $\left(2 + \frac{m}{8}\right) \times .45$

mpy: $\left(15 + .525m + \frac{m}{8}\right) \times .45$

div: $\left(20 + \frac{n}{4} + \frac{9}{4}\right) \times .45$